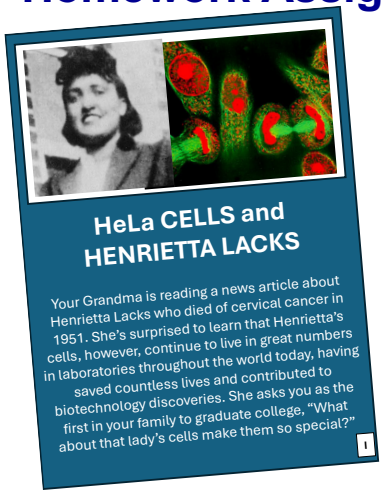
Biol 100: Class #6

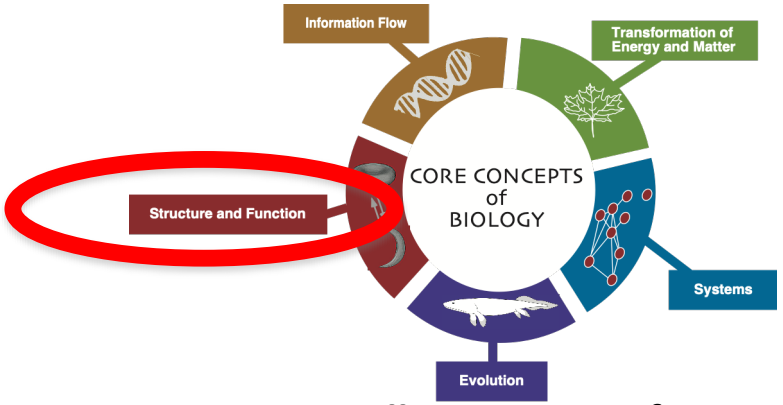
Agenda:

How do mutations in Proto-oncogenes & Tumor Suppressor genes cause cancer?
What is special about HeLa cells? Why is cancer rare in elephants?

- **Welcome and Agenda – EXAM #1 NEXT CLASS!!**
- **Sharing – 1) Human Biology in the News & 2) Strategies for Exam #1 Prep**
- **Finishing Three Big Ideas about Cancer – A Trait Caused by Gene Mutations**
  - **GAIN-of-Function Mutations in Proto-Oncogenes**
  - **LOSS-of-function Mutations in Tumor Suppressor Genes**
- **Applying What You Learned! Why is Cancer Rare in Elephants?**
- **Final Notes on… What is Different about HeLa Cells?**
- **Exam #1 Study Guide, Format, and Advice YOU CAN DO THIS!**
- **Passing the Baton to Sally – INFORMATION FLOW – a Few Clicker Questions…**
- **Homework Assignments – See Last Slide!**

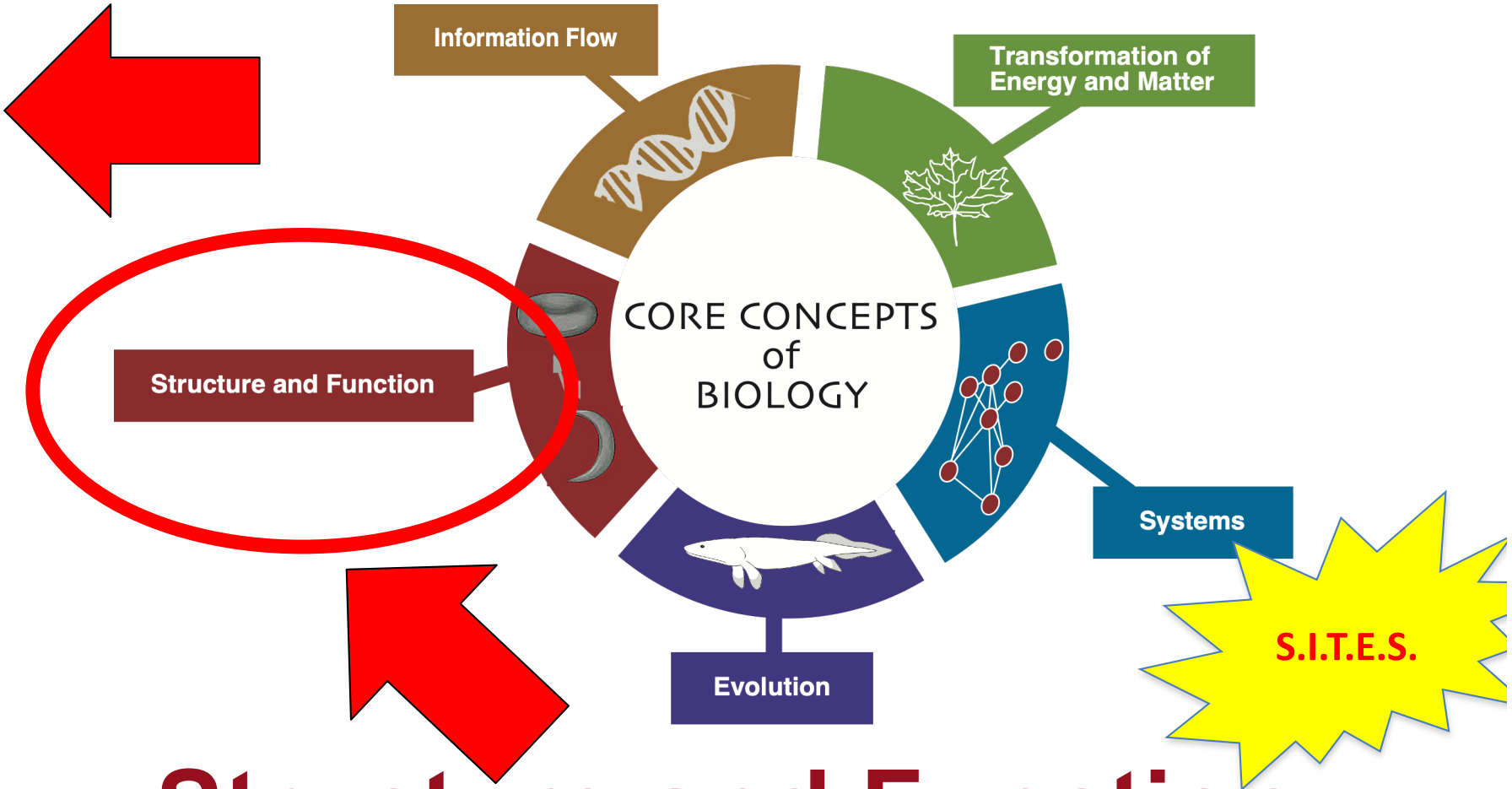






Sally Pasion, Professor
Kimberly Tanner, Professor





Structure and Function

(what something is made of) (what something can do)

Why should you care?!?

Thinking about the relationship between structure and function will help you learn and NOT memorize a bunch of biology terms…

Starting at the end… Exploring the Biological Mystery Boxes of Biol 100



THE ALTOIDS SCARE

Your best friend recently shared, "I don't eat mints any more after the Altoids scare. You offer, "Oh give me a break, what Altoids scare?"

She responds, "Well, I read on the internet that you can get Mad Cow Disease from Altoids. After all, they are made in Great Britain, and they do have gelatin in them."

How would you find out more and evaluate your friend's ideas?

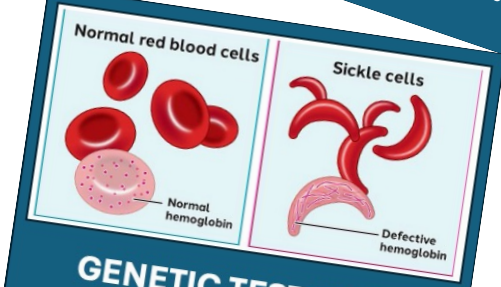
A



HeLa CELLS and HENRIETTA LACKS

Your Grandma is reading a news article about Henrietta Lacks who died of cervical cancer in 1951. She's surprised to learn that Henrietta's cells, however, continue to live in great numbers in laboratories throughout the world today, having saved countless lives and contributed to biotechnology discoveries. She asks you as the first in your family to graduate college, "What about that lady's cells make them so special?"

I



GENETIC TESTING & SICKLE CELL

In your new job at the Department of Public Health, your first appointment is with a pregnant woman named Maya. She has brought you her genetic testing results that screen for blood disorders like Sickle Cell Anemia and Thalassemia.

Maya shares, "There are lots of A, T, C, and G letters, but I'm confused. Could you help me understand?"

What do you know about DNA, genes, mutations, proteins, and traits?

And how could you help Maya understand?

E

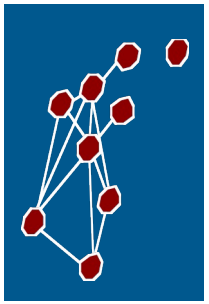


ELEPHANTS DON'T GET CANCER?

Despite their size and long life-span, only ~5% of elephants die from cancer, which is much less than for humans. Scientists recently discovered that elephants have 40 copies of the p53 gene, while humans have two copies (one from our biological mother and one from our biological father). Use the language of biology to explain how this discovery could explain why so few elephants die from cancer.

M

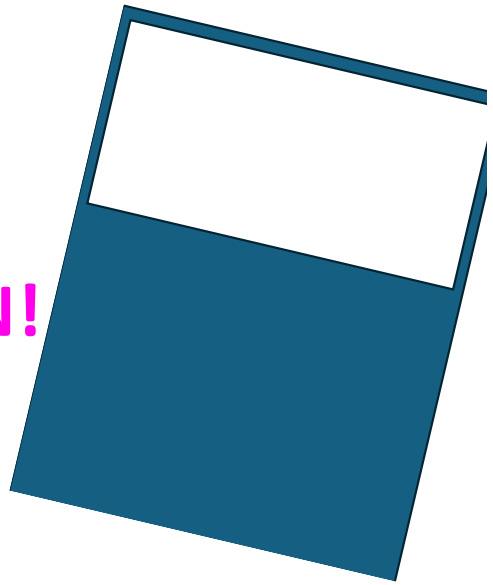



Sharing Your Human Biology in the News Explorations…

SHARE WITH COLLEAGUE NOW!

PREPARE TO SHARE on EXAM QUESTION!



Discussion Prompts:

1. What human biology topic and article did you choose? What was the source?
2. How does this topic connect Human Biology to your home community and/or career?

****

1. What **EVIDENCE** was referenced in the article?
2. What **QUESTIONS** did the article raise for you?
3. What are you **SKEPTICAL** about in the article?
4. What do you **WONDER** about the topic after having read the article?
5. What **CONFUSION** can you identify about your understanding of the topic at this point?

What will you remember later this week for Exam #1?

Thank you for your thoughtful reflective journal responses…



The Congressional Record

them with an educational environment that is conducive for productive learning. Among their scholastic accomplishments, this past year Key Largo School scored above average on the Stanford Achievement Test in Reading and Math, scored an average of 3.9 on the Florida Writes Exam with 96 percent of the eighth grade students scoring three or above, and, since 1972, has received consecutive accreditation by the Southern Association of Colleges and Schools. In addition, the school has written and received many large and small grants including a Customer Focused School Grant, Retrofit Grant, Home School Connection Grant, and Learn and Serve Grant. The grants have helped to make the school a model technology school for the district and the State; initiate the research, development and implementation of a block scheduling program; develop a theme based alternative education program for at risk students from grade 4 through 8; and service more than 300 exceptional students ranging with disabilities from pre-school handicapped to severely emotionally disturbed to gifted. The support of the community, business partners, teachers, and parents has been instrumental to the successful learning behaviors of the students at Key Largo Elementary and Middle School.

I commend Key Largo Elementary and Middle School on receiving the distinguished 1996–97 Blue Ribbon School Award. I know that the students and faculty will continue to exceed beyond their scholastic abilities and continue to be a model for schools throughout the country.

IN MEMORY OF HENRIETTA LACKS

HON. ROBERT L. EHRLICH, JR.

OF MARYLAND

IN THE HOUSE OF REPRESENTATIVES

Wednesday, June 4, 1997

Mr. EHRLICH. Mr. Speaker, I rise today to pay tribute to Henrietta Lacks, a woman whose contributions to medical science and research have gone relatively unnoticed for the past 46 years. Ms. Lacks provided a crucial sample of cells that has furthered our knowledge of medical science and disease prevention, and for this contribution, we are all grateful.

Henrietta Lacks was born in 1920 in Clover, VA. At the age of 23 she moved to Turner's Station, near Baltimore, MD, joining her husband David. She had five children, four of whom—Deborah, David Jr., Lawrence, and Zakariyya—still survive. Ms. Lacks was known as pleasant and smiling, and always willing to lend a helping hand.

After the birth of her fifth child, Ms. Lacks was admitted to the hospital at Johns Hopkins University where she was found to have cervical cancer. Before her death, she donated a tumor biopsy section which became the first human cell line to survive outside the body. This cell line has proven instrumental to medical research.

Due to traditional patient confidentiality requirements, Ms. Lacks was not acknowledged as the donor of the cells. Instead, the donor remained anonymous, and the cell line was known only as the HeLa cells. Under the care of Dr. George O. Gey, the cells flourished due to his innovative methods of preserving them. Dr. Gey went on to cultivate more cells which could be used for a variety of medical research. These cells proved instrumental in polio research, and they helped establish the fields of molecular biology and virology. Henrietta Lacks' cells are still used in research today, more than four decades after her death.

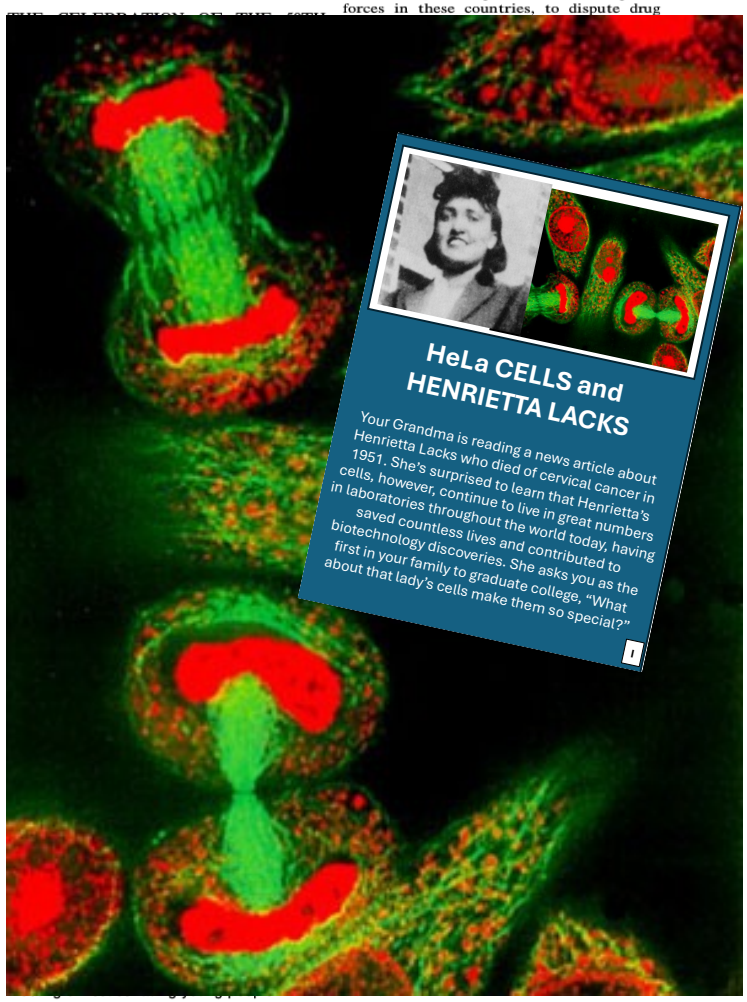
Henrietta Lacks' selfless contribution to the field of medicine has gone without acknowledgment for too long. Her cells made her immortal: through her death, countless others have been saved by the research that was made possible through her cell line. It is for this reason that I extend my deepest thanks to Henrietta Lacks and her family. I sincerely hope her name will also be immortalized as one of courage, hope, and strength, and that due recognition will be given to her role in medicine and science.

Mr. DOUTHAT. I would like to thank you for coming to our school, Congressman Sanders, and we are going to be talking about some of the U.S. drug policies in South America.

Clinton proposed to spend $16 billion this year on the War on Drugs. This figure is up from $10 billion spent during the 1980s over the entire decade. 70 percent of the money spent on drugs is spent on actual prevention and keeping it out of the country, whether that be through South American programs or Border Patrol and 30 percent is spent on drug rehabilitation.

During the Bush Administration he proposed a five-year, $2.2 billion program for Bolivia, Colombia, Ecuador and Peru, and in 1993 he granted $716 million to the South American nations and Mexico as long as they committed themselves to reducing their product production and respect for human rights.

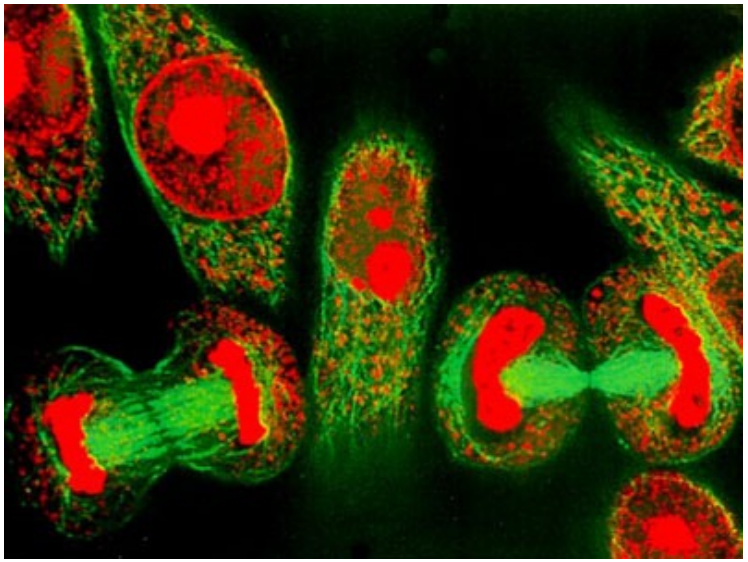
The U.S. also helped to train local police forces in these countries, to dispute drug





Share with a colleague…

What did you learn about the many ways that HeLa cancer cells are different from normal cells?





Excerpt from…

The Way of All Flesh

1997 Documentary

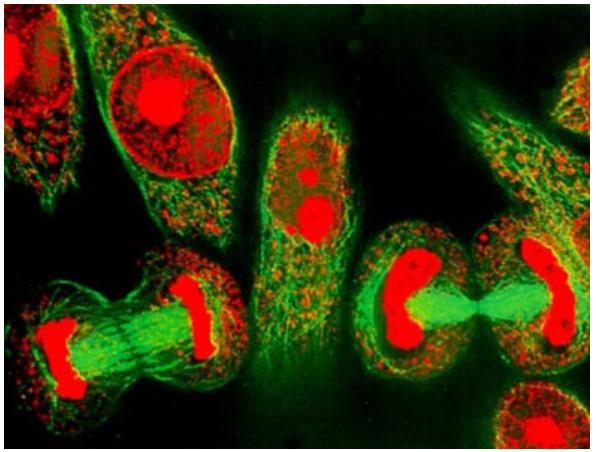
by British Filmmaker

Adam Curtis

Clip #1: 6:08-8:06

Feel free to watch more!

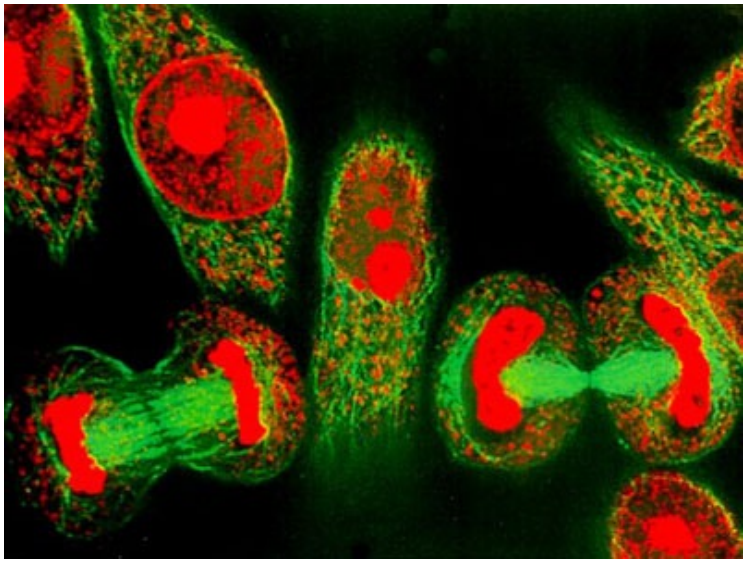
https://www.youtube.com/watch?v=C0lMrp_ySg8





Share with a colleague…

What was most interesting to you about the video excerpt?





Moving Towards Cancer…

Clicker Question

Thinking about yourself, your family, your friends, your colleagues and co-workers, *which of the following statements best describes your life experiences with cancer?*

A) I am a cancer survivor.

B) I have known a close friend or family member who had cancer in the past.

C) I have a close friend or family member who is struggling with cancer right now.

D) I have never known anyone personally who had cancer.

More than one of these might apply, but just pick one…



Understanding Cancer…

Clicker Question

Your friend was recently diagnosed with breast cancer. Your friend's breast cancer cells ***most likely*** have…

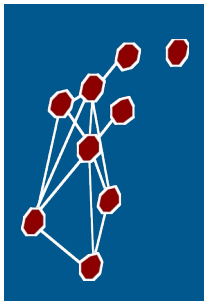
A. A single mutation in a gene that controls how cells divide

B. A silent mutation in a gene that controls how cells divide

C. Multiple mutations in genes that control how cells divide

D. Multiple mutations in genes that code for hemoglobin




Key Ideas about the Biology of Cancer

1. Cancer is Uncontrolled Cell Division (Mitosis)

2. Cancerous Cells Have Accumulated Genetic Damage through Mutations

3. At least TWO Mutation Events (and usually more) in Genes involved in Cells Dividing are Required for a Cell to Become Cancerous



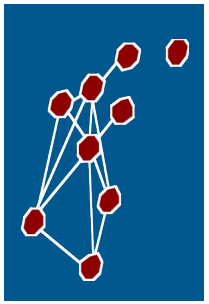
Understanding Cancer…

Clicker Question

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A. A single DNA mutation in a proto-oncogene.
B. A single DNA mutation in a tumor suppressor gene.
C. A silent DNA mutation in a tumor suppressor gene.
D. Multiple DNA mutations, with at least one in a proto-oncogene and one in a tumor suppressor gene.
E. Multiple DNA mutations, with at least one in a proto-oncogene and at least one in RNA polymerase.




Key Ideas about the Biology of Cancer

1. Cancer is Uncontrolled Cell Division (Mitosis)

2. Cancerous Cells Have Accumulated Genetic Damage through Mutations

3. At least TWO Mutation Events (and usually more) in Genes involved in Cells Dividing are Required for a Cell to Become Cancerous



Understanding Cancer…

Clicker Question

Based on your current understanding, which of the following things can cause MUTATIONS (changes) in GENES (in DNA) that can then result in CANCER?

A) Viruses

B) Ultraviolet radiation from the sun

C) Chemicals like those in tobacco

D) Only B and C

E) A, B, and C



So, what causes mutations–changes in GENES (in DNA) that code for traits?

DNA/Genes are "mutated" or "have mutations!"

This should NOT be said about cells…

Common Mutagens

(things that cause mutations in DNA)

- **UV (ultraviolet) radiation from sun**
- **X-Rays**
- **Variety of chemicals (benzene)**
- **Cigarette smoke**
- **Inhaled asbestos**
- **Viruses (like HPV in HeLa cells)**

…to name a few…

Spontaneous Mutations

(in the normal course of copying DNA when new cells are made)

Mutations are important in biology. Think evolution and variation!



Understanding Cancer…

Clicker Question

Based on your current understanding,

which of the following things can cause MUTATIONS in GENES that can then result in cancer?

A) Viruses

B) Ultraviolet radiation from the sun

C) Chemicals like those in tobacco

D) Only B and C

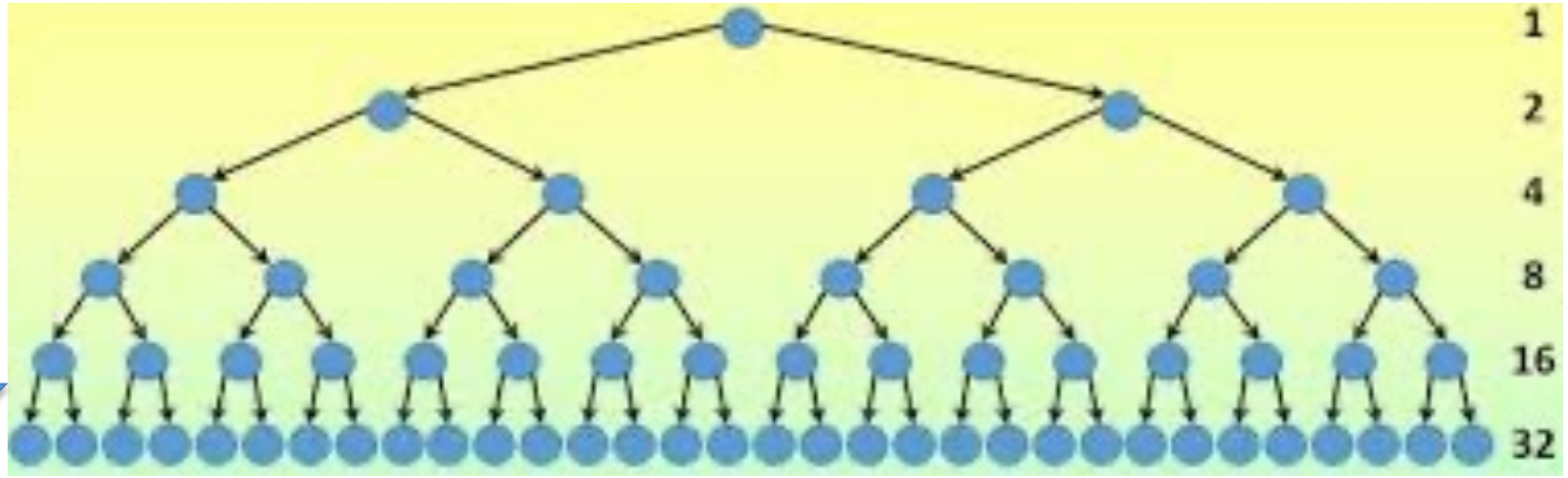
E) A, B, and C

1. Cancer is Uncontrolled Cell Division (Mitosis)



Human Skin Cell

TIME



Number of cells over time

division = one cell becoming two cells

"cells growing"

"cells dividing"

"cells multiplying"

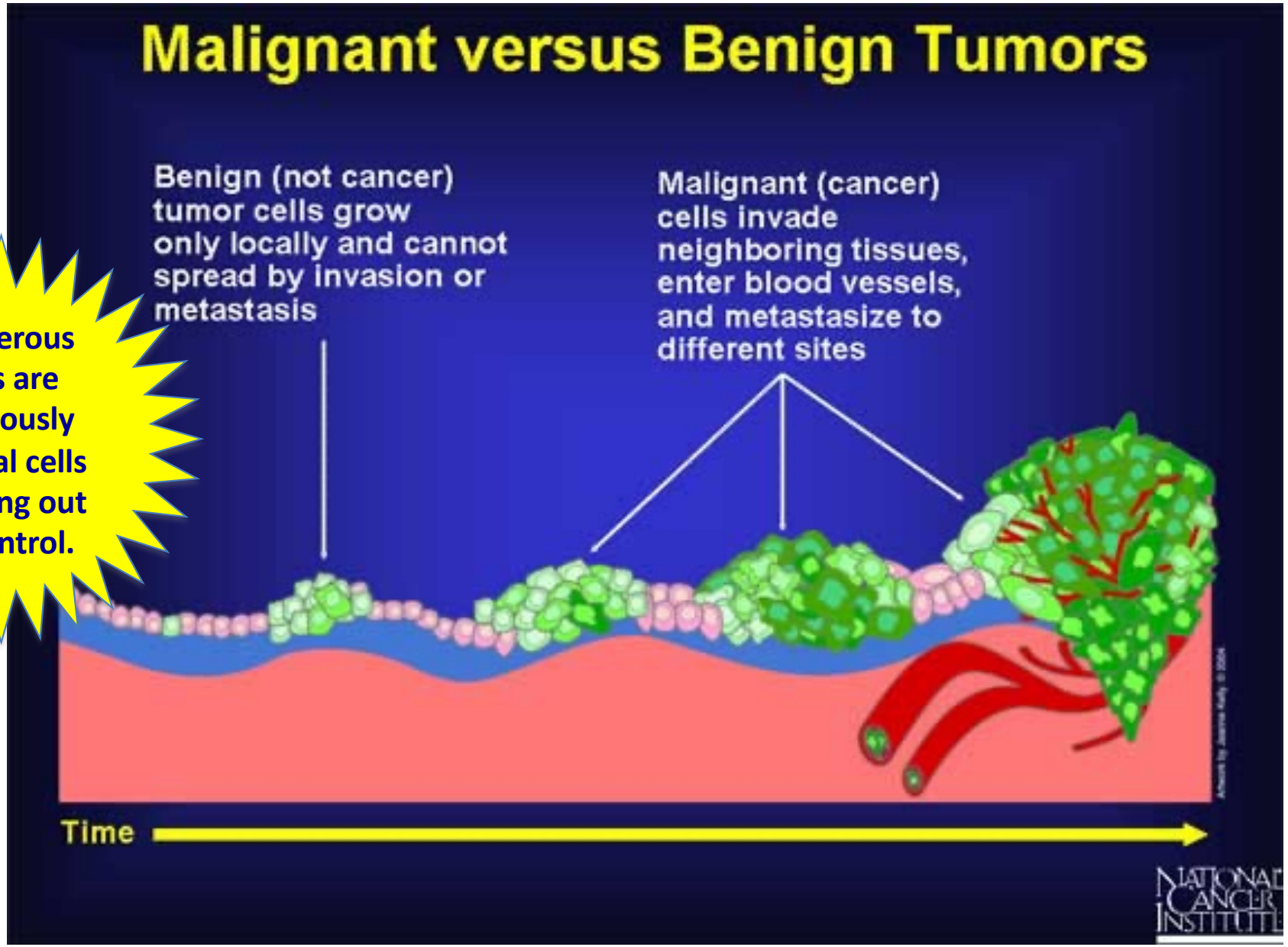
"cells copying themselves"

"out of control mitosis"

This is a common, everyday process in our cells…

NOTE: "growth" here does NOT mean one cell getting bigger

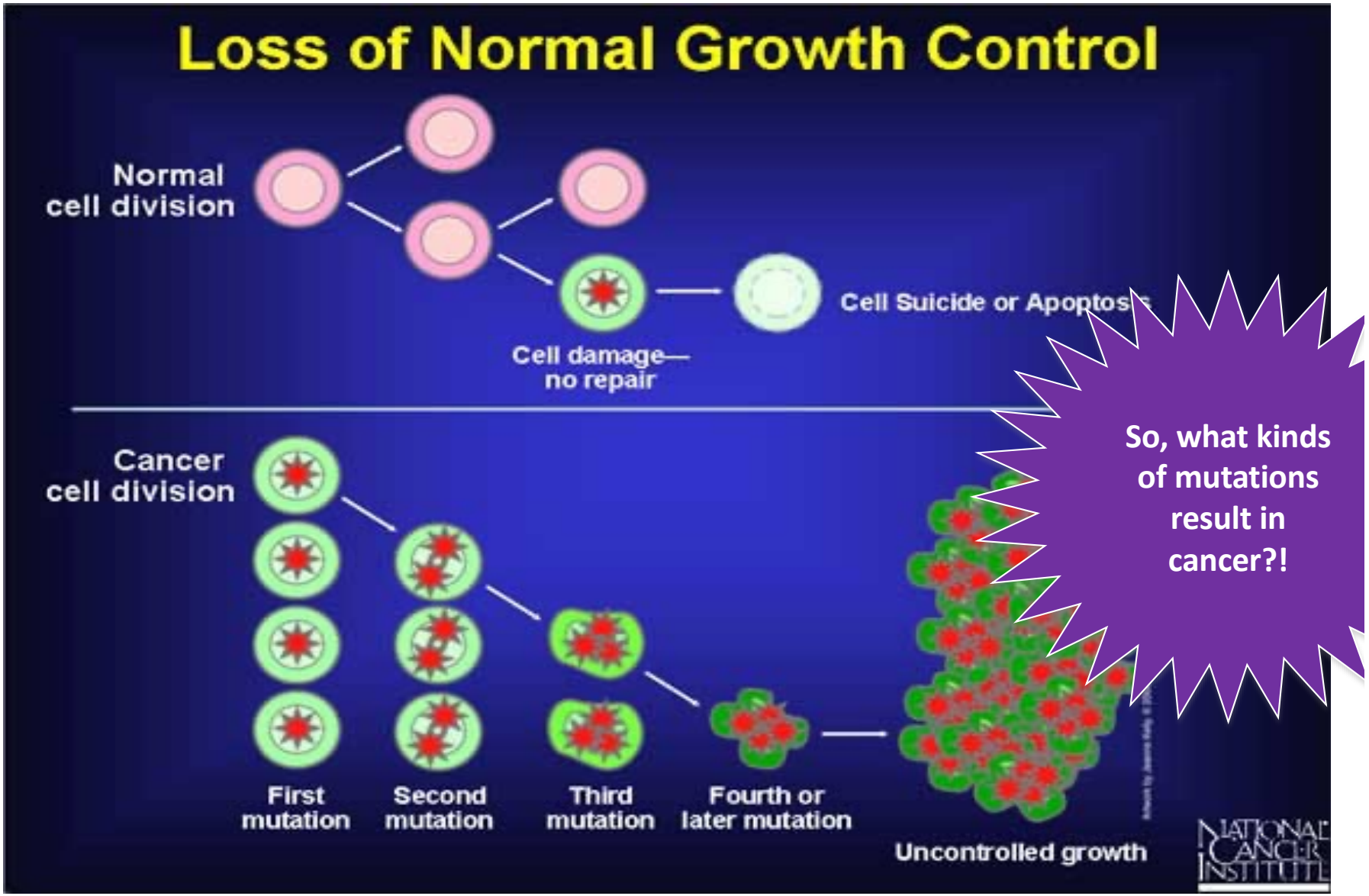
1. Cancer is Uncontrolled Cell Division (Mitosis)



Cancerous cells are previously normal cells growing out of control.

http://www.cancer.gov/cancertopics/cancerlibrary/what-is-cancer

2. Cancerous Cells Have Accumulated Genetic Damage through Mutations in Genes (DNA)



http://www.cancer.gov/cancertopics/cancerlibrary/what-is-cancer

3. At least TWO Mutation Events (and usually more) are Required for a Cell to Become Cancerous

An ACCELERATOR of cell division MUST get stuck!

AND

A BRAKE on cell division MUST be broken!

Analogy of controlling the speed of a car to controlling speed of cell division…



- Accelerator pedal (speed up cell division)
- Brake pedal (slow down cell division)

Both events have to happen for cell division (or car speed) to be out of control.

So, what kinds of mutations result in cancer?!



Understanding Cancer…

Clicker Question

The types of mutations that are hypothesized to underlie many types of cancer are….

A. gain-of-function mutations in tumor suppressor genes

B. loss-of-function mutations in tumor suppressor genes

C. gain-of-function mutations in proto-oncogenes

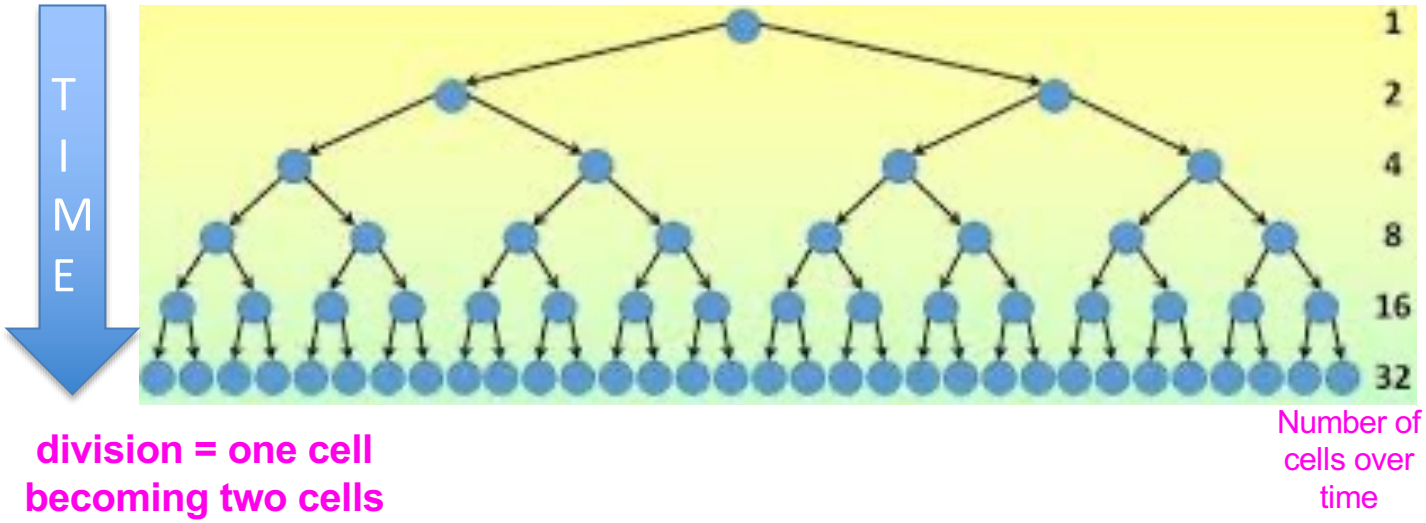
D. both A and B

E. both B and C

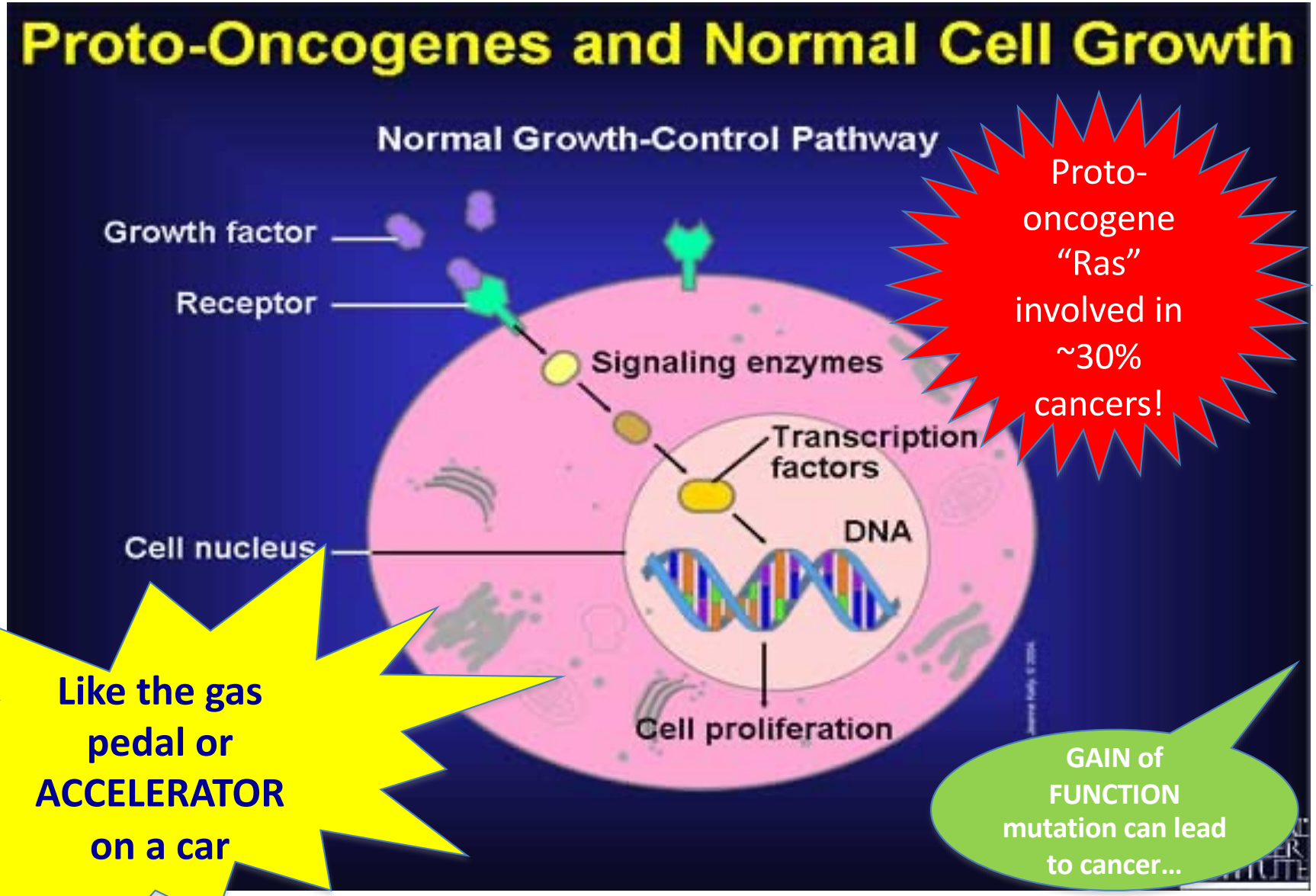
3. At least TWO Mutation Events (and usually more) are Required for a Cell to Become Cancerous

- Accelerator: Proto-Oncogenes and Proteins promote cell division
- Brake: Tumor Suppressor Genes and Proteins suppress cell division

These are normal genes that code for normal proteins in our cells that control speed of cell division!



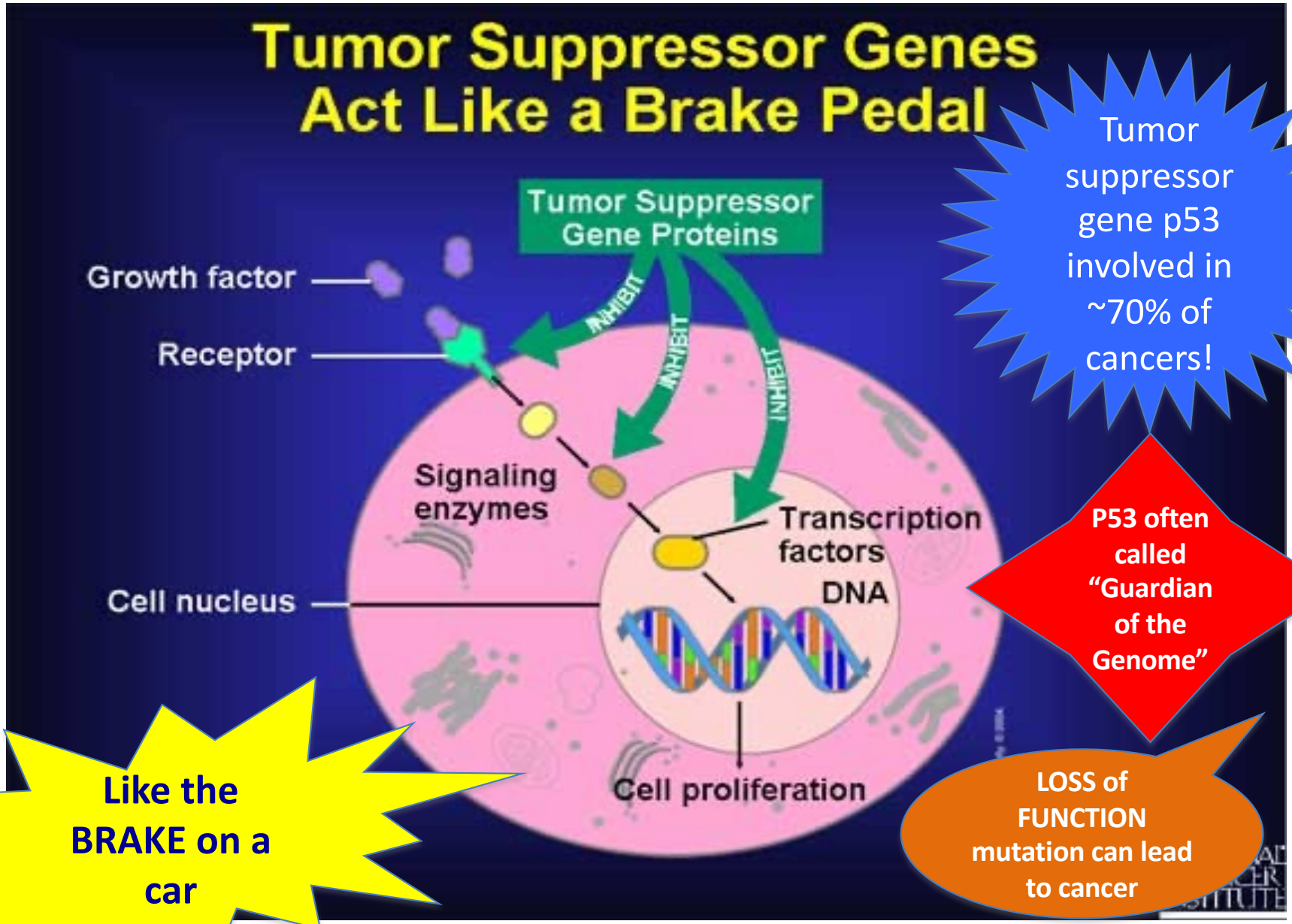
Proto-Oncogenes are Normal Genes that make Normal Proteins that Promote Cell Division



Like the gas pedal or ACCELERATOR on a car

http://www.cancer.gov/cancertopics

Tumor Suppressor Genes are Normal Genes that make Normal Proteins that Inhibit Abnormal Cell Growth



Tumor suppressor gene p53 involved in ~70% of cancers!

P53 often called "Guardian of the Genome"

LOSS of FUNCTION mutation can lead to cancer

Like the BRAKE on a car

http://www.cancer.gov/cancertopics



Remembering Types of Mutations…

Remember that single nucleotide mutations can be…

- *Substitution (silent produces same protein)*
- *Insertion (frameshift! always different protein)*
- *Deletion (frameshift! always different protein)*

ALSO, mutations that produce different proteins can either be…

"LOSS-of-FUNCTION" – protein can no longer do its job (function) at the same level or at all.

CANCER!

…but also (!)…

"GAIN-of-FUNCTION" – protein is produced that does something new or different (function) that it usually doesn't do.

3. At least TWO Mutation Events (and usually more) are Required for a Cell to Become Cancerous



An ACCELERATOR of cell division MUST get stuck!

GAIN of Function Mutation in a PROTO-ONCOGENE

AND



AND

A BRAKE on cell division MUST be broken!

LOSS of Function Mutation in a TUMOR SUPPRESSOR GENE



Understanding Cancer…

Clicker Question

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B. loss-of-function mutations in tumor suppressor genes

C. gain-of-function mutations in proto-oncogenes

D. both A and B

E. both B and C



Understanding Cancer…

Clicker Question

The types of mutations that are hypothesized to underlie many types of cancer are….

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B. loss-of-function mutations in tumor suppressor genes

C. loss-of-function mutations in proto-oncogenes

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E. both B and C



Understanding Cancer…

Clicker Question

Your friend was recently diagnosed with breast cancer. Your friend's breast cancer cells ***most likely*** have…

A. A single DNA mutation in a proto-oncogene.
B. A single DNA mutation in a tumor suppressor gene.
C. A silent DNA mutation in a tumor suppressor gene.
D. Multiple DNA mutations, with at least one in a proto-oncogene and one in a tumor suppressor gene.
E. Multiple DNA mutations, with at least one in a proto-oncogene and at least one in RNA polymerase.

3. At least TWO Mutation Events (and usually more) are Required for a Cell to Become Cancerous



An ACCELERATOR of cell division MUST get stuck!

GAIN of Function Mutation in a PROTO-ONCOGENE (ras)

AND



CANCER

Good overview here:
http://www.cancer.gov/cancertopics

AND

A BRAKE on cell division MUST be broken!

LOSS of Function Mutation in a TUMOR SUPPRESSOR GENE (p53)



Clicker Question!

You are reading a science fiction novel about a biotechnologically advanced society where individuals must choose one of the following types of mutations for a future child to have in all its cells before it is born.

Which of the following mutations would you choose for your future child, and why?

A. Gain of function mutation in a proto-oncogene
B. Gain of function mutation in the p53 gene
C. Loss of function mutation in tumor suppressor gene
D. Deletion mutation in the p53 gene
E. Insertion mutation in a tumor suppressor gene

Exploring Elephants and Cancer…



ELEPHANTS DON'T GET CANCER?

Despite their size and long life-span, only ~5% of elephants die from cancer, which is much less than for humans. Scientists recently discovered that elephants have 40 copies of the p53 gene, while humans have two copies (one from our biological mother and one from our biological father). Use the language of biology to explain how this discovery could explain why so few elephants die from cancer.

M

GAIN of Function Mutation in a PROTO-ONCOGENE

!CANCER!

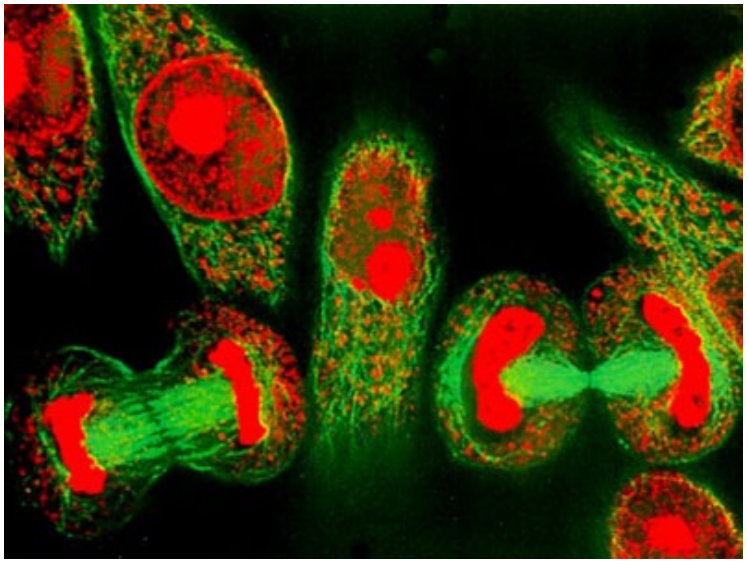
LOSS of Function Mutation in a TUMOR SUPPRESSOR GENE (p53)

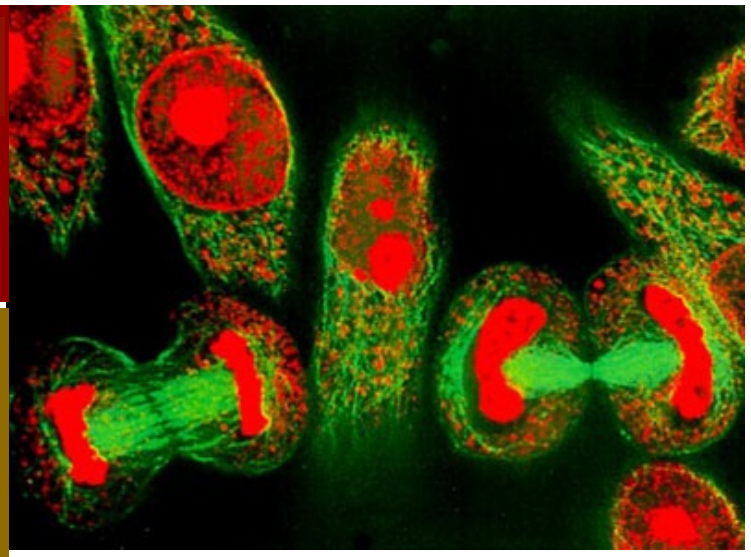
Using what we've just explored about cancer, how would you explain why elephants are less likely to have cancer?



Share with a colleague…

What did you learn about the many ways that HeLa cancer cells are different from normal cells?





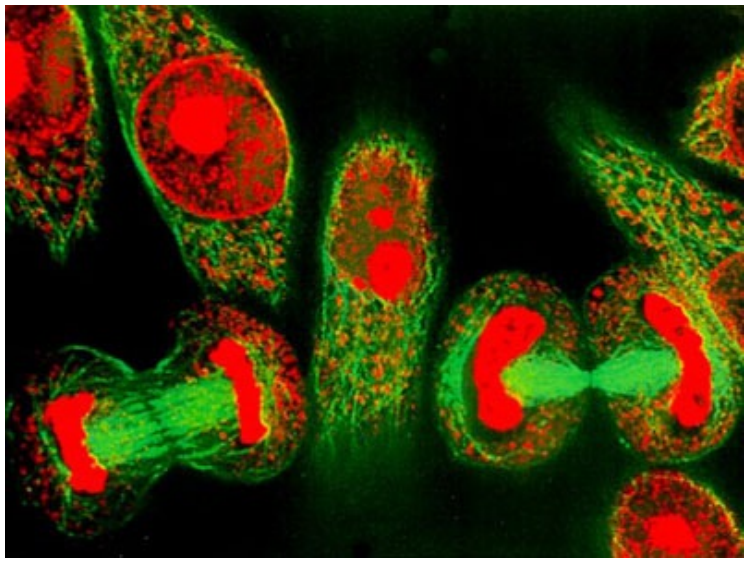
About what is known about why HeLa cells are so special…

I'm curious if there's something unique to Henrietta Lacks' cells… If it is the case that her cells are specifically capable of this, then what in Ms. Lacks' DNA gave her cancerous cells this ability?

They were the first immortalized human cell line. They were instrumental in development of the polio vaccine. They have been used in in tens of thousands of research publications and have earned two Nobel Prizes.

My burning question is: "What makes HeLa cells so special?"…I did a little poking around on the internet and found that HeLa DNA has 82 chromosomes, not 46… and it has several kinds of genes from the human papilloma virus associated with cervical cancer. It replicates abnormally fast and has an active copy of telomerase, which makes it replicate indefinitely.

Why do HeLa cells have these super-human immortal powers?" A good explanation came in the 90's when human cancer cells have an enzyme called telomerase that builds the ends of telomeres on DNA, preventing aging. a researcher discovered that when normal cells divide their telomeres get a little shorter until old age when their telomeres are so short it prevents the cells from further dividing. It leads me to the question can we somehow effectively deliver telomerase to our cells we could could live forever?!



About what is known about why HeLa cells are so special…

I'm curious if there's something unique to Henrietta Lacks' cells… If it is the case that her cells are specifically capable of this, then what in Ms. Lacks' DNA gave her cancerous cells this ability?

Some things that make HeLa cells different…

…have many more chromosomes (~80!) than normal human body cells (usually 46)

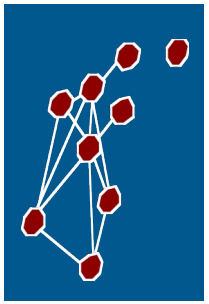
…have mutations caused by human papilloma virus (HPV) (both GAIN-of-Function in Proto-Oncogenes and LOSS-of-Function in Tumor Suppressor Genes)

…have active Telomerase protein that allows damaged cells to keep dividing



Go Team Biol 100!




Information about Format of Exam #1…

Exam #1 is…

- During Class #7, **NEXT CLASS! THIS WEEK!**
- No materials needed, except something to write with
- Half multiple choice, half open-ended short answer and essay questions
- Extra Credit Questions – 10 points!
- No use of notes or electronic devices

Because we value Team Biol 100 collaboration and we see exams as learning opportunities…

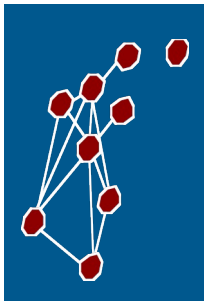
First 50 minutes of Class #7

- Take the exam independently, using a black or blue ink pen

Last 25 minutes of Class #7

- Collaborate with any colleagues to check your answers and make corrections in your exam paper using a special pen.
  - can correct multiple choice with new answer and sentence explaining why you changed your mind
  - can update your short essays with additional information on the back of the page
- **Only resources for corrections are your Team Biol 100 colleagues!**




Name Tent Reflection…

What are the 6 habits of mind we focus on practicing in this Human Biology course?

1._____

2._____

3._____

4._____

5._____

6._____

U-ABC-IT!





Clicker Question

Which of the following human cells contains the <u>genes that determine eye color</u>?

A. Only cells in the liver

B. Only gametes (egg or sperm)

C. Only cells in the eye

D. Gametes and cells in the eye, but not cells in the liver

E. Gametes, cells in the eye, and cells in the liver



Clicker Question

What of the following is ordered from LARGEST to SMALLEST?

A. mitochondrion, hemoglobin, virus, glucose, water molecule

B. virus, mitochondrion, hemoglobin, glucose, water molecule

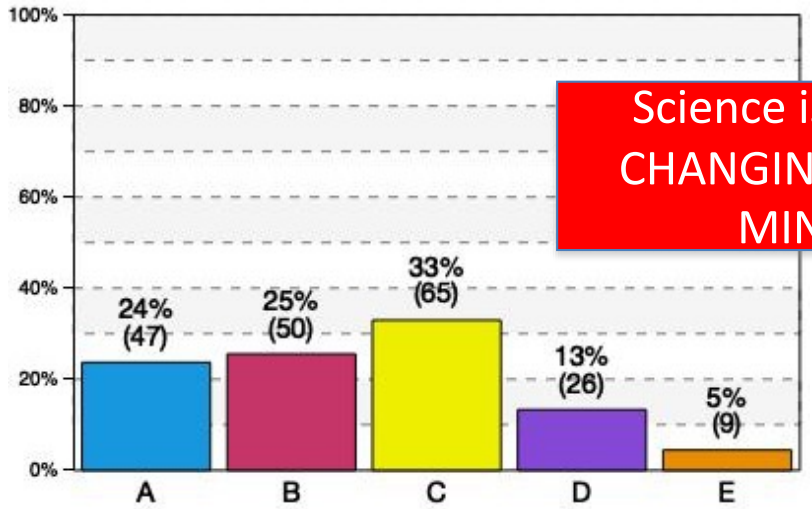
C. mitochondrion, virus, hemoglobin, glucose, water molecule

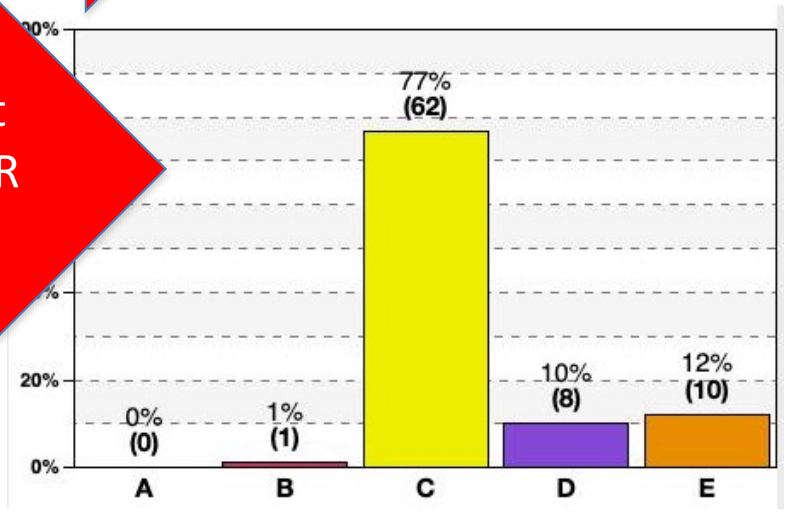
D. virus, mitochondrion, glucose, hemoglobin, water molecule

E. I really have no idea…

Hopefully, learning!

Science is about CHANGING YOUR MIND

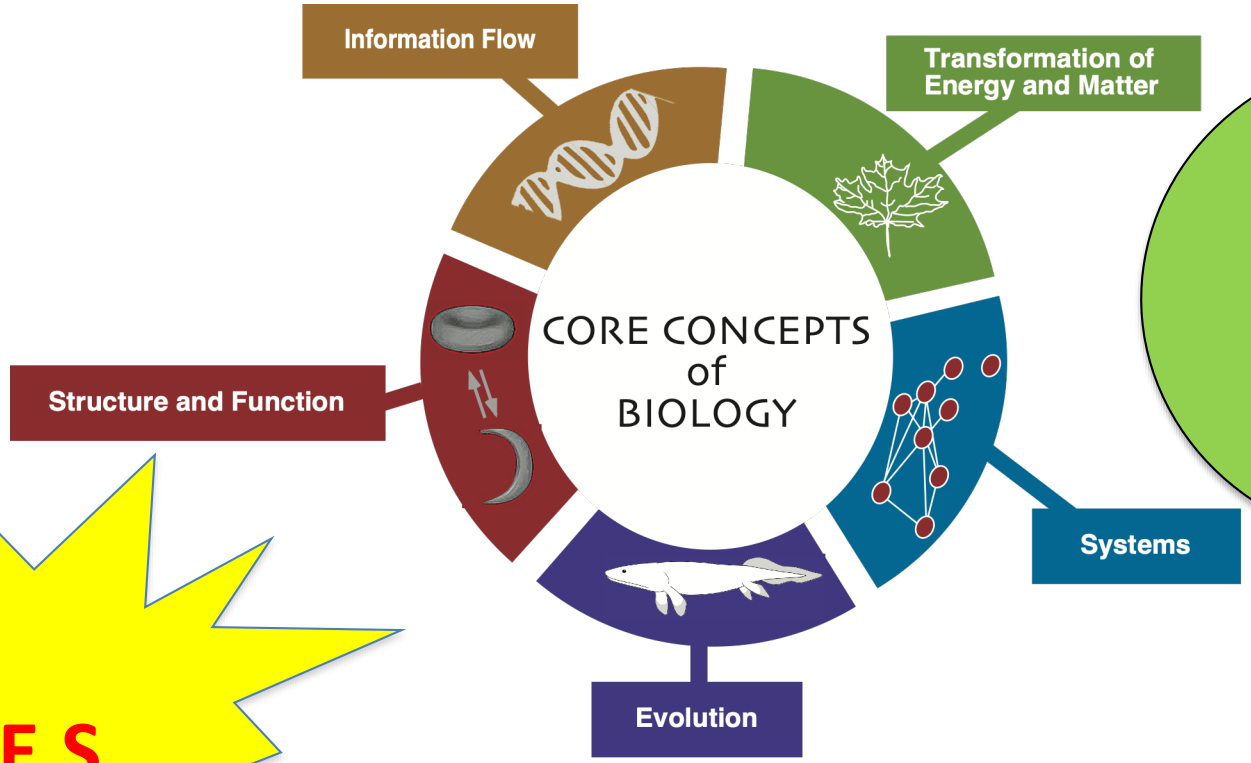




Passing of the Baton!



Big Idea: Five Core Concepts of Biology!



It's about ORGANIZING YOUR KNOWLEDGE (not just memorizing)

S.I.T.E.S.

Structure and Function

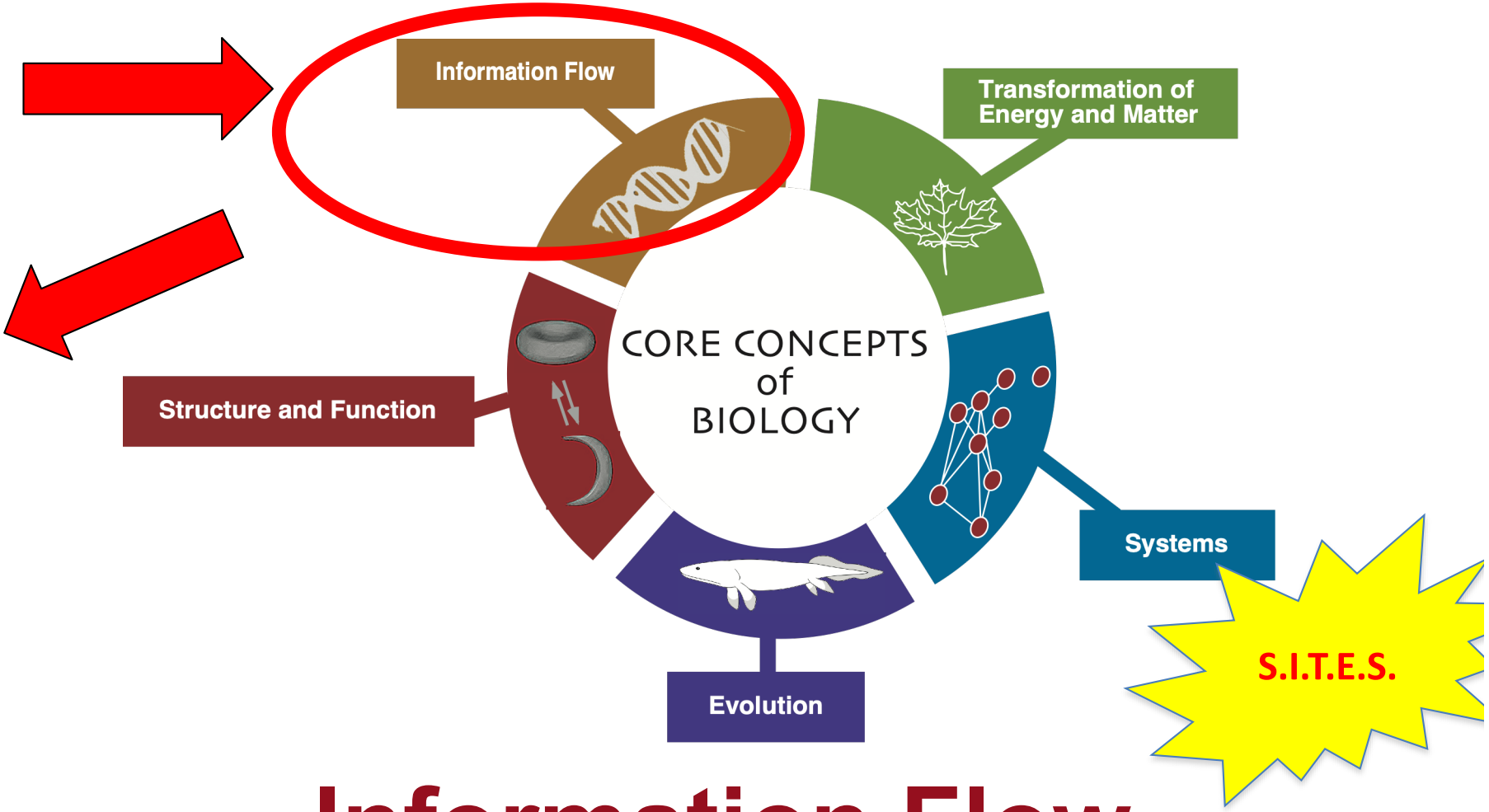
Information Flow

Transformations of Energy & Matter

Evolution

Systems Biology





Information Flow

(how gene codes are passed from cell to cell AND generation to generation)

Why should you care?!?

Thinking about information flow can help you understand where you come from (biologically) and what you will leave behind when you are gone (biologically)…



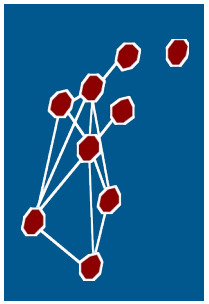
Clicker Question
FOR AFTER EXAM #1!

Choose the answer that best describes your response to the statement:

“Peanut butter can cause cancer.”

A. Yes, peanut butter is always linked to cancer in humans.
B. Yes, peanut butter can be linked to cancer in humans.
C. No, peanut butter is never linked to cancer in humans.
D. I have no idea, and I love peanut butter.
E. I have no idea, and I hate peanut butter.




Homework! FOR AFTER EXAM #1

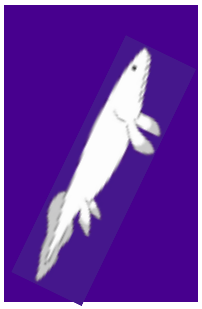
Due by midnight (11:59pm) Sunday, 9/20 on Canvas

- **Reflective Journal #5: The Peanut Files!**
- **See instructions and link to the Peanut Files website on Canvas Week 5**
- **Please SCROLL DOWN when you arrive at the LINK to access lots of materials…**

Go Team Bio 100!







Clicker Question
FOR AFTER EXAM #1!

A young man develops skin cancer on his left shoulder that does not spread to any other tissues.

The mutation responsible for the cancer arose in a single cell in his skin.

If he and a female partner (who does not have skin cancer) have children after his skin cancer diagnosis, which of the following is most likely true?

A. All the man's children will inherit the mutation

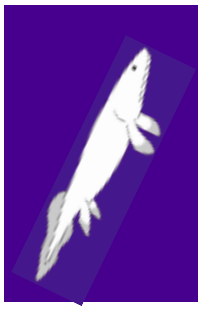
B. All the man's children will inherit the mutation if the mutation is dominant

C. Some of the man's children may inherit the mutation depending on which of his chromosomes they inherit

D. None of the man's children will inherit the mutation





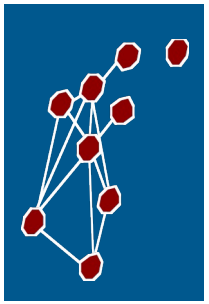


Clicker Question
FOR AFTER EXAM #1!

Approximately what percentage of all cancer diagnoses worldwide are primarily caused by inherited genetic mutations passed down from a parent?

A. All the time
B. Most of the time
C. Half of the time
D. Occasionally
E. Never




Assignments

EXAM #1 NEXT CLASS!!! Study, Study, Study!

Due Sunday, September 20 by Midnight (11:59pm)

- **Complete Reflective Journal #5 – Instructions! Reflective Journal #5-The Peanut Files on Canvas. PLEASE NOTE! You will need to SCROLL DOWN on the Peanut Files website to access materials…**

Open from Sunday, September 20 to Saturday, September 26 at Midnight (11:59pm)

- **Self-Assessment Quiz #4, see Canvas**
- **Also, please use the PRACTICE versions of previous Self-Assessment Quizzes on Canvas to study.**

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REMINDER! Homework is graded as on time during the 48-hour grace period after the posted deadline. Homework submitted after the grace period is not graded.